

Ashir Rashid

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EDUCATION

New York University Abu Dhabi

B.S. in Computer Science; Minor in Applied Mathematics

Abu Dhabi, UAE

Estimated Graduation: May 2026

Relevant Courses

Algorithmic Foundations of Data Science, Network Security, Foundations of ML, NLP, Algorithms, Data Structures, Data Management and Analysis, Operating Systems, Computer Networking, Probability and Statistics, Linear Algebra

Certifications

[AWS Certified Cloud Practitioner](#)

Jan 2025 - Jan 2028

PUBLICATIONS

"Rethinking Evaluation of Multiple Sclerosis (MS) Lesion Segmentation Models" A. Basit, A. Rashid, M. A. Hanif, M. Shafique Accepted at *IEEE IJCNN @ WCCI 2026*

WORK EXPERIENCE

GamaLearn

Data Scientist

Abu Dhabi, UAE

May 2025 - May 2026

- Designed scalable ELT pipelines using Airflow to process **100,000+** records, enabling robust statistical modeling.
- Developed automated analytical **dashboards** to monitor Key Performance Indicators (**KPIs**) in near real-time, reducing reporting latency from ~1 hour to minutes and improving decision-making efficiency.
- Developed and rigorously **backtested** probabilistic models, including EM-based models, Gaussian Mixture Models, and Bayesian estimators using variational inference and MCMC to evaluate robustness and distributional stability.
- Improved out-of-sample predictive accuracy by **25%** and reduced Expected Calibration Error (**ECE**) by **30%** through model refinement and distribution-aware parameter optimization.

eBrain Lab

Deep Learning Research Assistant

Abu Dhabi, UAE

Aug 2023 - May 2025

- Built targeted synthetic datasets across **FLAIR, T1, and T2** modalities for **Minimum Functionality, Invariance, and Directional Expectation** tests to expose blind spots in existing losses and rigorously evaluate dataset coverage.
- Integrated **advanced loss functions**, including Generalized Surface Loss, to overcome the fundamental weaknesses of standard boundary-based and region-based segmentation losses, improving balanced accuracy by ~7%.
- Engineered a flexible per-segment **evaluation framework** that applies metrics and losses at the lesion level, closing critical gaps in GSL and enabling far more precise **model diagnostics**.
- Enhanced energy and memory efficiency of **Deep Neural Networks** by **8.3%**, including **Large Language Models**, through implementing a state-of-the-art quantization framework, expanding on an existing open framework (HAQ).
- Optimized quantization configurations by applying **non-linear optimization** methods (**RL, Evolutionary Search, and DNAS**), improving model performance by **15%** and reducing hyperparameter search time by **43%**.
- Streamlined experiments with an **MLFlow-based MLOps** pipeline, reducing experiment iteration time by **12%**.

Gantom Lighting & Controls

Full Stack Engineer

California, US (Remote)

Jan 2022 - Jul 2023

- Built and deployed a scalable cloud-based web platform on **Google Cloud Platform** using Django and Firestore, managing **hundreds** of product entities with optimized data retrieval, increasing user engagement by **28%**.
- Applied database optimization and caching strategies to reduce latency by **22%**.

OPEN-SOURCE EXPERIENCE

IntelOwl

Open-source OSINT solution for gathering threat intelligence data

Jan 2023 - Present

- Fortified phishing defense capabilities by integrating the CheckPhish **API** with the dynamic React **front-end**.
- Rectified **system-level integration** and **CI/CD** bottlenecks, improving scalability and collaborative development.

PERSONAL PROJECTS

Source code for these projects is available on my [GitHub](#)

- Gmail RAG - LangChain, LLMs: End-to-end RAG pipeline for Gmail data, enabling semantic search and retrieval.
- Better RAG - Elasticsearch, HPC: Using Information Extraction techniques to improve RAG retrieval accuracy.
- P2P File Sharing - Python, Sockets: A file transfer app utilizing a peer-to-peer approach, similar to BitTorrent.

SKILLS

- Programming: Python, C++, C, SQL, PyTorch, TensorFlow, NumPy, Scikit-learn
- Modeling & Optimization: Bayesian Inference, MCMC, Variational Inference, EM, Gaussian Mixture Models, Reinforcement Learning, Evolutionary Search, Constrained Optimization
- Systems & Infrastructure: Airflow, MLflow, Docker, Kubernetes, AWS, GCP, PostgreSQL, Elasticsearch, HPC